Supplementary Material S3. Concise Study-Level Taxonomy of the Included Studies (n = 71)

"No." refers to the record number in S1; "Ref." refers to the reference number in the paper. Rows are ordered by reference number.

Database	No.	Ref.	Author(s)	Application area	Prediction task	Model(s)	XAI method(s)	Performance metrics
Scopus	72	8	Farooq et al. (2026)	Transportation / Vehicle fuel consumption and CO_2 emissions	Fuel consumption and CO_2 emission prediction	Linear Regression (LR), Ridge Regression (Ridge), Least Absolute Shrinkage and Selection Operator (Lasso), Random Forest (RF), AdaBoost Regression (AdaBoost), Gradient Boosting (GB), Extra Trees Regressor (ETR), Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), Support Vector Regression (SVR)	LIME (Local)	R^2, RMSE, MAE, MSE
Scopus	70	9	Ranjan and Madanayak (2026)	Transportation / Freight Operations	CO_2 emission intensity prediction	Multiple Linear Regression (MLR), Bayesian Ridge Regression (BRR), LASSO, Random Forest (RF), XGBoost (XGB)	SHAP (global & local), Partial Dependence Plots	R^2, RMSE, MAE, MSE
Scopus	110	10	Şahin (2025)	Maritime transportation / Passenger vessels / Carbon emissions	Hourly CO_2 emissions prediction from passenger ships	Random Forest (RF), Extreme Gradient Boosting (XGBoost)	SHAP (Global)	R^2, RMSE, MAE, Relative MAE (%)
Scopus	42	11	Özüpak (2026)	Aviation CO_2 emissions / Sustainable aviation	Monthly aviation CO_2 emissions prediction	Physics-Based Model (PBM), Long Short-Term Memory (LSTM), One-Dimensional Convolutional Neural Network (1D-CNN), Transformer Network (Transformer), Hybrid Physics-Guided Model (HPGM), Stacked Ensemble Model (Ensemble)	SHAP (Global + Local), LIME (Local)	R^2, RMSE, MAE, NSE, Theil U2, Accuracy, Precision, Recall, F1-score, AUC
Scopus	50	12	Gungor et al. (2026)	Building energy performance / Campus CO_2 emissions	Building-level annual CO_2 emissions prediction	Support Vector Regression (SVR), Bayesian-Optimized Support Vector Regression (SVR_BO), Gaussian Process Regression (GPR), Least-Squares Boosting (LSBoost), Physics-Informed LSBoost (LSBoost_PI), Artificial Neural Network (ANN), Bagged Trees (Bagged), Regression Tree (Tree), Ridge Regression (Ridge), Stacking Ensemble (Stacking)	SHAP (Global + Local), Permutation Importance (Global), Ablation Analysis (Global)	RMSE, MSE, MAE, MedAE, Bias, R^2, MAPE, NRMSE, EVS, Corr, RMSLE
Scopus	32	13	Wani et al. (2026)	Construction materials / Sustainable concrete	CO_2 emissions prediction and minimization in GGBFS-based geopolymer concrete	XGBoost (XGB), LightGBM (LGBM), CatBoost (CB), AdaBoost (AB), Artificial Neural Network (ANN)	SHAP (Global + Local), LIME (Local), PDP	R^2, MAE, RMSE, MSE, NSE, MAPE
Scopus	16	14	Zhu et al. (2026)	Urban carbon emissions / Nighttime-light-based estimation / Regional mitigation	City-level carbon emission prediction and heterogeneous driver analysis in the Yangtze River Delta	Decision Tree (DT), Random Forest (RF), Extreme Gradient Boosting (XGBoost), Support Vector Machine (SVM), K-Nearest Neighbors (KNN)	SHAP (Global + Local), PDP (Global + Interaction)	R^2, MSE, MAE
Scopus	23	15	Song et al. (2026)	Agriculture / Dairy cattle slurry / GHG emissions	CH_4 emission prediction	LR, RF, XGBoost (XGB), LightGBM (LGBM), SVR, MLP	SHAP (Global + Local)	R^2, RMSE, MAE
Scopus	29	16	Song et al. (2026)	Power systems / Carbon management	Substation carbon-emission factor (CEF) prediction	Random Forest (RF), Gradient Boosting Machine (GBM), Long Short-Term Memory (LSTM), Pure Transformer, MOIRAI-2 + UPINN	Any-Variate Attention (Global + Local), UPINN Physical Decomposition (Global + Local), Counterfactual Reasoning (Local)	MAE, RMSE, R^2
Scopus	192	17	Assael et al. (2023)	Corporate sustainability / Greenhouse gas emissions / Carbon accounting	Estimation of non-reported corporate Scope 1 and Scope 2 GHG emissions	Light Gradient Boosting Machine (LightGBM)	SHAP (Global + Local)	R^2, RMSE, MAE

Database	No.	Ref.	Author(s)	Application area	Prediction task	Model(s)	XAI method(s)	Performance metrics
Scopus	108	18	Zhou et al. (2025)	Carbon markets / Carbon pricing / Investment decision support	One- and three-step-ahead carbon price point and interval forecasting	Integrated Carbon Price Analysis Framework (DNMADE); Grid Search–Echo State Network (GS–ESN); Grid Search–Backpropagation Neural Network (GS–BPnn); Grid Search–Long Short-Term Memory (GS–LSTM); Grid Search–Convolutional Neural Network–Long Short-Term Memory (GS–CNNLSTM); Grid Search–Transformer (GS–Transformer); Grid Search–Decomposition Linear Model (GS–DLinear); Multi-objective Grasshopper Optimization Algorithm–Outlier-Robust Extreme Learning Machine (MGOA–ORELM); Cosine Whale Optimization Algorithm–Extreme Learning Machine (CWOA–ELM); Sparrow Search Algorithm–Backpropagation Neural Network (SSA–BPnn); Slime Mold Algorithm–Bidirectional Long Short-Term Memory (SMA–BiLSTM); Hybrid Kernel Density Estimation (HKDE); Lower–Upper Bound Estimation (LUBE); Quantile Temporal Convolutional Network (QTCN); Gaussian Process Regression (GPR)	AW (Global + Local), SHAP (Global + Local)	MAE, RMSE, MAPE, MSPE, IA; PICP, PINAW, AIS, CWC; DA
Scopus	158	19	Liang et al. (2025)	Carbon markets / Emissions trading / Carbon pricing	Carbon price forecasting in China's Hubei carbon market	Linear Regression (LR), Light Gradient Boosting Machine (LGBM), Artificial Neural Network (ANN), K-Nearest Neighbors (KNN), Long Short-Term Memory (LSTM), and Autoregressive Integrated Moving Average (ARIMA), used within EEMD-based hybrid frameworks; Ensemble Empirical Mode Decomposition–Generalized Additive Interpretable Neural Network–Long Short-Term Memory–Autoregressive Integrated Moving Average (EEMD–GAMI_NET–LSTM–ARIMA)	ALE (Global)	R^2, RMSE, MAE, DA
Scopus	4	28	Ye et al. (2026)	Transportation / Urban transportation systems / Carbon emissions	Transportation carbon emission (TCE) prediction	Linear Regression (LR), Random Forest (RF), XGBoost, LightGBM, CatBoost	SHAP (Global + Local)	R^2, RMSE, MAE, MAPE
Scopus	2	29	Özüpak and Aslan (2027)	Greenhouse gas emissions / Climate change	Long-term CO_2 and N_2O emissions prediction	Decision Tree (DT), K-Nearest Neighbors (KNN), AdaBoost, Support Vector Machine (SVM), Optimized Random Forest (RF)	SHAP (Global + Local), LIME (Local)	R^2, RMSE, MAE, MSE, Max Error
Scopus	41	30	Yang et al. (2026)	Carbon markets / Regional ETS / Carbon pricing	Daily carbon price forecasting across China's eight ETS pilots	Explainable Carbon Price Forecasting Network (XCPFNet; CEEMDAN–CNN–LSTM–Transformer); Empirical Mode Decomposition–Temporal Information–Long Short-Term Memory (EMD–TI–LSTM); Complete Ensemble Empirical Mode Decomposition with Adaptive Noise–Long Short-Term Memory–Transformer (CEEMDAN–LSTM–Transformer); Complete Ensemble Empirical Mode Decomposition with Adaptive Noise–Support Vector Regression–Long Short-Term Memory (CEEMDAN–SVR–LSTM)	Clustered SHAP (Global + Local + Interaction); Kernel SHAP (Global + Local; comparison baseline)	MSE, MAE, RMSE, MAPE, R^2; PIPC, PINAW; ES, FIC, TS

Database	No.	Ref.	Author(s)	Application area	Prediction task	Model(s)	XAI method(s)	Performance metrics
Scopus	1	31	Ali et al. (2027)	Agriculture / Agri-food systems / Carbon management	CO_2 emission prediction	Random Forest Regressor (RF)	LIME (Local), SHAP (Global), DiCE (Counterfactual)	R^2, MSE, RMSE, MAE, MAPE, Median Absolute Error, Max Error, Explained Variance Score
Scopus	8	32	Saraswat et al. (2026)	Transportation systems / Vehicle emissions	Vehicle CO_2 emissions prediction	Multi-Layer Perceptron (MLP), GAO-MLP (MLGO), HLOA-MLP (MLHO), XGBoost (XGB), Random Forest (RFR), SVR	SHAP (Global + Local), RFE, CAM	RMSE, R^2, MAPE, MARE, TIC
Scopus	27	33	Shaik et al. (2026)	Transportation systems / Vehicle emissions	Vehicular CO_2 emissions prediction	Halfway Escape Optimization–TabNet (HEO-TabNet), Particle Swarm Optimization–TabNet (PSO-TabNet), Firefighter Optimization–TabNet (FFO-TabNet)	TabNet Feature Masks (Global + Local), SHAP (Global + Local), LIME (Local)	RMSE, MAE, R^2, MAPE
Scopus	90	34	Saleem et al. (2026)	Intelligent transportation / CAVs	Fuel consumption and CO_2 emission prediction	Extra Trees, Linear Regression, XGBoost, Bayesian Ridge, Ridge Regression, Gradient Boosting, Random Forest (RF), Decision Tree (DT), AdaBoost, Lasso Regression, Support Vector Regression (SVR-RBF), K-Nearest Neighbors (KNN) Regressor, ElasticNet, LightGBM, Huber Regressor	LIME (Local), SHAP (Global)	R^2, RMSE, MAE
Scopus	109	35	Alam et al. (2025)	Transportation / Vehicle emissions / Environmental sustainability	Vehicle CO_2 emission prediction	Decision Tree Regressor (DT), K-Nearest Neighbors (KNN), Support Vector Regression (SVR), Extreme Gradient Boosting (XGBoost), AdaBoost Regressor (AdaBoost), CatBoost Regressor (CatBoost), Ensemble (DT + KNN + XGBoost), Long Short-Term Memory (LSTM), Bidirectional Long Short-Term Memory (BiLSTM), Carbon Multilayer Perceptron (CarbonMLP)	SHAP (Global + Local)	R^2, MSE, RMSE, MAPE
Scopus	131	36	Yuan et al. (2025)	Transportation / Vehicle emissions / Sustainable transport	ICE and HEV CO_2 emissions prediction	Linear Regression (LR), Decision Tree (DT), Random Forest (RF), Extreme Gradient Boosting (XGBoost)	ALE (Global), Feature Importance Analysis (Global)	R^2, MSE, RMSE
Scopus	167	37	Guo et al. (2024)	Transportation / Vehicle emissions	Vehicle CO_2 emission prediction	K-Nearest Neighbors (KNN), Support Vector Regression (SVR), Gradient Boosting Machine (GBM), Decision Tree (DT), Random Forest (RF), Lasso Regression (LASSO)	SHAP (Global), ALE (Global)	R^2, RMSE, rRMSE, MAE, MAPE
Scopus	33	38	Luo et al. (2026)	Transportation systems / Tunnel CO_2 emissions	Instantaneous vehicle CO_2 emissions prediction	Exponential Vehicle Specific Power Model (VSP), Multiple Linear Regression (MLR), Random Forest (RF), eXtreme Gradient Boosting (XGBoost)	SHAP (Global), ICE (Local)	R^2, RMSE
Scopus	28	39	Chen et al. (2026)	Transportation systems / Transportation carbon emissions	Transportation CO_2 emissions prediction and multi-scenario forecasting	Extra Trees (ET)	SHAP (Global + Local)	MSE, RMSE, R^2
Scopus	116	40	Huang et al. (2025)	Urban transportation / Neighborhood-level planning	Neighborhood-level road traffic CO_2 emission prediction	Extreme Gradient Boosting (XGBoost)	SHAP (Global + Local), GAM (Nonlinear effects)	R^2, RMSE, MAE
Scopus	133	41	Zhao and He (2025)	Transportation / Urban transportation systems / Carbon emissions	City-level transportation carbon emission prediction	Categorical Boosting (CatBoost), Light Gradient Boosting Machine (LightGBM), Extreme Gradient Boosting (XGBoost), Random Forest (RF), Gradient Boosted Regression (GBR), Multilayer Perceptron (MLP), Extreme Learning Machine (ELM), Bayesian Regression (Bayes), K-Nearest Neighbors (KNN), Decision Tree (DT)	SHAP (Global + Local + Interaction)	R^2, RMSE, MAE, MAPE

Database	No.	Ref.	Author(s)	Application area	Prediction task	Model(s)	XAI method(s)	Performance metrics
Scopus	137	42	Roy et al. (2025)	Transportation / Urban traffic / Signalised intersections	Joint prediction of traffic CO_2 emissions and total travel delay at signalised intersections	Support Vector Regression (SVR), Random Forest (RF), Extreme Gradient Boosting (XGBoost), Gradient Boosting (GB), Multi-task Convolutional Neural Network (MTCNN), Multi-task Temporal Convolutional Network (MT2CN)	SHAP (Global + Interaction)	R^2, RMSE, MAE, MAPE
WoS only	70	43	Liu et al. (2025)	Transportation / Public transportation / Carbon emissions	Metro-bus combined travel (MBCT) CO_2 emission reduction rate prediction	Gradient Boosting Decision Tree (GBDT), Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), Categorical Boosting (CatBoost), Ordinary Least Squares (OLS)	SHAP (Global + Local), PDP (Global)	R^2, RMSE, MAE
Scopus	73	44	Fang et al. (2026)	Residential buildings / Building operational carbon emissions	Hourly operational carbon emission prediction	Autoregressive Model (AR), Decision Tree (DT), Gradient Boosting Decision Tree (GBDT), Support Vector Machine (SVM), Feedforward Neural Network (FNN), Gaussian Process Regression (GPR), Sageformer, Temporal Pattern Attention Long Short-Term Memory (TPA-LSTM), Multivariate Time-Series Graph Neural Network (MT-GNN), Dynamic Correlation Graph–Temporal Convolutional Network–Graph Convolutional Network (DCG-TCN-GCN)	SHAP (Global + Local)	Corr, RSE, RMSE
Scopus	114	45	Zhang et al. (2025)	Campus / Buildings / Community-scale carbon management	CO_2 emissions prediction and optimization for a university campus	Random Forest Regressor (RF)	SHAP (Global), Pearson Correlation Analysis (Global)	R^2, RMSE
Scopus	193	46	Zhang et al. (2023)	Buildings / Building energy efficiency / GHG emissions	Residential building energy consumption and GHG emission prediction	Light Gradient Boosting Machine (LightGBM), Extreme Gradient Boosting (XGBoost), Random Forest (RF), Support Vector Regression (SVR)	SHAP (Global + Local), PDP (Global)	R^2, MSE, RMSE, MAE
WoS only	75	47	Zhou et al. (2024)	Buildings / Office buildings / Carbon emissions / Energy efficiency	Building carbon emission, photovoltaic (PV) carbon offset, and net carbon emission prediction	Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM)	SHAP (Global + Local + Interaction)	R^2, MAPE, MAE
Scopus	180	48	Khan et al. (2024)	Green buildings / Sustainable construction / Building energy performance	Energy consumption, CO_2 emissions, and thermal discomfort prediction	Light Gradient Boosting Machine (LGBM), Bayesian Optimization–LightGBM (BO-LGBM), Bayesian Optimization–XGBoost (BO-XGB)	LIME (Local)	R^2, MAPE
Scopus	61	49	Cao et al. (2026)	Construction materials / Steam-cured concrete	Carbon emission and concrete performance prediction in steam-curing processes	Random Forest (RF), BO-RF, XGBoost, BO-XGBoost, CatBoost, BO-CatBoost; improved NSGA-III (optimization)	SHAP (Global &Local)	R^2, RMSE, MAE
Scopus	59	50	Wang et al. (2026)	Construction / Carbon emissions	Carbon emission prediction	Linear Regression (LR), Random Forest (RF), Backpropagation Neural Network (BPNN), Extreme Gradient Boosting (XGB)	SHAP (global), PDP (feature effects)	R^2, MAE, RMSE, MAPE
Scopus	51	51	Seo et al. (2026)	Construction / Built Environment	Carbon emission prediction (construction equipment, pre-construction stage, regression)	Decision Tree (DT), Random Forest (RF), XGBoost (XGB), Support Vector Regression (SVR), Multi-Layer Perceptron (MLP), Ensemble (DT+SVR etc.)	SHAP (Global + Local + Interaction)	MAE, SMAPE
Scopus	88	52	Zheng and Shuang (2026)	Building sector / Embodied carbon emissions	Embodied carbon emissions (BEC) prediction	Support Vector Machine (SVM), Artificial Neural Network (ANN), Random Forest (RF), Extreme Gradient Boosting (XGBoost), Categorical Boosting (CatBoost), Back-Propagation Neural Network (BP), Long Short-Term Memory (LSTM)	SHAP (Global + Local)	MAE, MAPE, RMSE, R^2

Database	No.	Ref.	Author(s)	Application area	Prediction task	Model(s)	XAI method(s)	Performance metrics
Scopus	98	53	Zawaideh (2026)	High-rise construction / BIM-based risk management / Embodied carbon	Cost overrun, schedule delay, and embodied-carbon prediction	Extreme Gradient Boosting (XGBoost), Random Forest (RF), Light Gradient Boosting Machine (LightGBM), Categorical Boosting (CatBoost), Support Vector Regression (SVR), Artificial Neural Network (ANN)	SHAP (Global)	R^2, RMSE, MAE, MAPE
Scopus	115	54	Liu et al. (2025)	Civil engineering / Bridge construction / Carbon emissions	Bridge construction CO_2 emission prediction	Random Forest (RF), Gradient Boosting Regression Trees (GBRT), Categorical Boosting (CatBoost), Extreme Gradient Boosting (XGBoost), Adaptive Boosting (AdaBoost), Multilayer Perceptron (MLP), Decision Tree Regression (DT), Support Vector Regression (SVR)	SHAP (Global + Local), PDP (Global), ICE (Local)	R^2, MSE, MAE, RMSE, Composite Weighted Index (CWI)
Scopus	17	55	Chen et al. (2026)	Urban sustainability / Carbon emissions	Urban carbon emissions prediction	Support Vector Regression (SVR), Random Forest (RF), Light Gradient Boosting Machine (LightGBM), AdaBoost, CatBoost, eXtreme Gradient Boosting (XGBoost)	SHAP (Global + Local)	R^2, MSE, RMSE, MAE
Scopus	94	56	Ren et al. (2026)	Urban sustainability / Urban morphology / Carbon management	Sectoral and total CO_2 emission classification and prediction	ResNet-18 (CNN), XGBoost, Mediation–Moderation Regression	Grad-CAM (Spatial / Local), SHAP (Global + Local)	Accuracy, Precision, R^2, MAE, RMSE, EVS
Scopus	95	57	Bhowmik et al. (2026)	Forest monitoring / Climate mitigation / Land-use change	Tree cover loss and CO_2 emissions forecasting	Linear Regression (LR), Random Forest (RF), XGBoost, Long Short-Term Memory (LSTM)	SHAP (Global)	R^2, RMSE, MAE
Scopus	99	58	Jiang et al. (2026)	Land use / Carbon emissions and storage / Regional carbon management	County-level land-use carbon emission forecasting	Extreme Gradient Boosting (XGBoost), Random Forest (RF), Patch-generating Land Use Simulation (PLUS), Cellular Automata–Markov (CA–Markov), Integrated Valuation of Ecosystem Services and Tradeoffs (InVEST), Carbon Emission Coefficient Method	SHAP (Global + Local + Interaction)	R^2, Kappa
Scopus	130	59	Bilotta et al. (2025)	Urban sustainability / Smart cities / Carbon emissions	Fine-scale urban CO_2 emission estimation	Linear Regression (LR) using Ordinary Least Squares (OLS), Extreme Gradient Boosting (XGBoost), Graph Convolutional Network (GCN)	SHAP (Global + Local)	R^2, MSE, RMSE, MAE, MAPE
Scopus	132	60	Zhang et al. (2025)	Urban sustainability / Regional carbon emissions / Spatiotemporal analysis	Fine-scale energy-related carbon emission estimation and spatiotemporal reconstruction	Linear Regression (LR), Support Vector Regression (SVR), Random Forest Regression (RFR), Extreme Gradient Boosting (XGBoost)	SHAP (Global + Local + Interaction)	R^2, RMSE, MAE, Prediction Relative Deviation (PRD)
Scopus	152	61	Xu et al. (2025)	Urban planning / Residential land carbon emissions	Residential land carbon emission prediction	XGBoost, Gradient Boosting Decision Tree (GBDT), Random Forest (RF), AdaBoost, Logistic model	SHAP (Global + Local + Interaction)	R^2, MAE, MSE, RMSE, MAD, MAPE, EVS, MSLE
Scopus	155	62	Wang et al. (2025)	Urban sustainability / Digital transformation	Carbon emission intensity (CEI) prediction	OLS Regression, Ridge Regression, LASSO Regression, Gradient Boosting Decision Tree (GBDT), Random Forest (RF), XGBoost, Light Gradient Boosting Machine (LGBM)	SHAP (Global, Local, Interaction)	R^2, RMSE, MAE
Scopus	185	63	Luo et al. (2024)	Land use / Regional sustainability / Carbon emissions	Multi-scale land-use-based CO_2 emission prediction	Linear Regression (LR), Multilayer Perceptron (MLP), K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Optimized Extra Trees Regressor (ETR), Random Forest (RF), Decision Tree (DT), Extreme Gradient Boosting (XGBoost), Optimized Land Expansion Analysis Strategy (Opti-LEAS), Cellular Automaton based on Multiple Random Seeds (CARS), Markov Chain (MC)	SHAP (Global + Local)	R^2, MSE, RMSE, MRE, MAE, MBE

Database	No.	Ref.	Author(s)	Application area	Prediction task	Model(s)	XAI method(s)	Performance metrics
Scopus	191	64	Yao et al. (2023)	Urban carbon emissions / Urban sustainability	CO_2 emission turning-point forecasting and identification of socioeconomic drivers	Environmental Kuznets Curve (EKC), Autoregressive Integrated Moving Average (ARIMA), Error Correction Model (ECM), Decision Tree (DT), Genetic Algorithm-optimized EKC (GA-EKC), Stepwise Regression (SR), Extreme Gradient Boosting (XGBoost)	SHAP (Global + Local)	R^2, Adjusted R^2
Scopus	5	65	Sireesha et al. (2026)	Greenhouse gas emissions / Industrial and energy sectors	Multi-gas GHG emission prediction (CO_2, CH_4, N_2O, F-gases)	Gradient Boosting (GB), Random Forest (RF), XGBoost, CatBoost, AdaBoost	SHAP (Global + Local), LIME (Local), ICE, PDP	R^2, RMSE, MAE, VAF, Explained Variance, WI, PBIAS
Scopus	74	66	Dahmani et al. (2026)	Climate change / Environmental forecasting	Simultaneous CO_2 emission and temperature prediction	Conv1D-LSTM (Multi-Task Learning)	LIME (Local)	R^2, RMSE, MAE, MSE
Scopus	76	67	Ghannam (2026)	National CO_2 emissions / Climate policy	Australia CO_2 emissions forecasting	AutoRegressive Integrated Moving Average (ARIMA), Random Forest (RF), XGBoost, Support Vector Regression (SVR), Gaussian Process Regression (GPR), Extreme Learning Machine (ELM), Improved Sparrow Search Algorithm–Extreme Learning Machine (ISSA-ELM), Multi-Layer Perceptron (MLP), Long Short-Term Memory (LSTM), Recurrent Neural Network (RNN), Gated Recurrent Unit (GRU), Stacking Regressor, Enhanced Stacking Regressor	SHAP (Global), Tree-based feature importance (Global)	MSE, RMSE, R^2, MAE, MAPE, MSLE, MedAE
Scopus	87	68	Ma et al. (2026)	Climate change / Carbon emissions	CO_2 emissions prediction	Random Forest (RF), XGBoost, LightGBM, CatBoost, Multilayer Perceptron (MLP)	SHAP (Global + Local)	R^2, MSE, RMSE, MAE, MAPE
Scopus	151	69	Zheng and Shuang (2025)	Climate / Country-level carbon emissions	Carbon emission, peaking and neutrality prediction	Support Vector Machine (SVM), Artificial Neural Network (ANN), Random Forest (RF), XGBoost, CatBoost, Back-Propagation Neural Network (BP), Long Short-Term Memory (LSTM)	SHAP (Global + Local)	MAE, RMSE, MAPE, R^2
Scopus	153	70	Han and Lin (2025)	Environmental economics / Global country-level CO_2 emissions	Long-run CO_2 emission forecasting and Environmental Kuznets Curve analysis	Linear Regression (LR), Random Forest (RF), Support Vector Regression (SVR), XGBoost (XGB), Artificial Neural Network (ANN)	SHAP (Global + Local)	MAE, MSE, RMSE, R^2
Scopus	156	71	Chen et al. (2025)	Regional carbon management / Provincial emissions / Emission-reduction pathways	Provincial carbon emission prediction; scenario-based projections for 2030 and 2060	Extended Belief Rule Base (EBRB), Cumulative Belief Rule Base (CBRB), Cumulative Belief Rule Base with Interpretability Criteria (CBRB-IC), Autoregressive Integrated Moving Average (ARIMA), Grey Model (GM), Extreme Gradient Boosting (XGBoost), Belief Rule Base (BRB), Data Envelopment Analysis–Extended Belief Rule Base (DEA-EBRB), Bi-level Joint Optimization–Extended Belief Rule Base (Bi-EBRB)	SHAP (Global + Local); intrinsic rule-based interpretability	MAE, MSE, RMSE, MAPE, R; modeling time, inference time, rule count, antecedent attribute count
Scopus	187	72	Mangla et al. (2024)	Energy economics / Energy risk management / Global GHG emissions	Total GHG emissions prediction and identification of key emission drivers	Random Forest (RF), Multilayer Perceptron (MLP), Deep Neural Network (DNN)	SHAP (Global + Local)	R^2, Adjusted R^2 (regression validation); Support, Confidence, Lift (association rules)

Database	No.	Ref.	Author(s)	Application area	Prediction task	Model(s)	XAI method(s)	Performance metrics
Scopus	203	73	Crespo et al. (2021)	Regional/sectoral carbon emissions / Environmental policy	Total and sector-specific EU28 CO_2-emission prediction	Backpropagation Artificial Neural Network (NN/BP), Neural Network with Backward Feature Elimination (NN/BP-BFE), Neural Network with RReliefF-driven Backward Feature Elimination (NN/BP-RReliefF/BFE), RReliefF-driven Iteratively Optimized Backpropagation Neural Network (Proposed Framework), Gradient Boosting Machine with Backward Feature Elimination (GBM-BFE), Random Forest with Backward Feature Elimination (RF-BFE), Support Vector Machine with Backward Feature Elimination (SVM-BFE)	LIME (Local), RReliefF (Global Feature Importance)	RMSE, MAPE
Scopus	54	74	Shi et al. (2026)	Energy / Urban sustainability / Carbon emissions	Energy-related CO_2 emission estimation and spatial reconstruction	Linear Regression (LR), Random Forest Regression (RFR), Support Vector Regression (SVR), Extreme Gradient Boosting (XGBoost)	SHAP (Global + Interaction)	R^2, RMSE
Scopus	75	75	Qader and Albalooshi (2026)	Energy systems / Power-station CO_2 emissions	Power-station CO_2 emissions forecasting	Extreme Gradient Boosting (XGBoost), Random Forest (RF), Long Short-Term Memory (LSTM), Support Vector Regression (SVR)	SHAP (Global + Local), LIME (Local)	RMSE, MAE, R^2
Scopus	78	76	Ramalingam et al. (2026)	Sustainable biofuel combustion / Engine performance and emission prediction	Biofuel combustion performance and emission prediction	Extreme Gradient Boosting (XGBoost), Adaptive Boosting (AdaBoost), Gradient Boosting Machine (GBM)	SHAP (Global + Local), Feature Importance Analysis (Global)	R^2, RMSE, MSE, MAE
Scopus	100	77	Sayghe et al. (2026)	Power systems / Energy forecasting / Smart grid	Multi-task forecasting of electricity load, prices, CO_2 emissions, and renewable generation	Autoregressive Integrated Moving Average (ARIMA), Prophet, Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Temporal Convolutional Network (TCN), Temporal Fusion Transformer (TFT), Neural Basis Expansion Analysis for Time Series (N-BEATS), Informer, Autoformer, Patch Time Series Transformer (PatchTST), TimesFM, Chronos, Moirai, Mixture-of-Experts Moirai (Moirai-MoE), Fine-Tuned TimesFM (TimesFM-FT), Fine-Tuned Chronos (Chronos-FT), Fine-Tuned Moirai-MoE (Moirai-MoE-FT), Grid Foundation Model (GridFM)	SHAP (Global + Local), Attention Visualization (Global)	MAPE, RMSE, MAE, sMAPE, CRPS, Calibration metrics
Scopus	71	78	Sireesha & Sheik (2026)	Agriculture sector / GHG emissions	CO_2, CH_4, N_2O emissions prediction	Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), Gated Recurrent Unit (GRU), Deep Feedforward Neural Network (DFNN)	SHAP (Global + Local), LIME, ICE, PDP	RMSE, MAE, R^2, VAF
Scopus	157	79	Jeaunita et al. (2025)	Precision Agriculture / Smart Farming	CO_2 emission prediction / Anomaly detection	Feedforward Neural Network (FNN), Deep Autoencoder, Random Forest (RF), Long Short-Term Memory (LSTM), Linear Regression (LR), Support Vector Regression (SVR), Shallow FNN, 1D CNN, Bi-GRU, Shallow Autoencoder, Deep DNN	SHAP (Global), LIME (Local)	R^2, MSE, RMSE, Accuracy, Precision, Recall, F1 Score
Scopus	45	80	Chai et al. (2026)	Enterprise carbon footprint tracking / Sustainable enterprises	Near-real-time Scope 1–2 emissions prediction and reduction optimization	Transformer (Transformer), Convolutional Neural Network (CNN), Bidirectional Long Short-Term Memory (BiLSTM), Ensemble Deep Learning Model (Ensemble: Transformer + CNN + BiLSTM)	SHAP (Global + Local + Interaction), LIME (Local)	MAPE, RMSE, Pearson r, Emission reduction (%), ROI, NPV, Payback period

Database	No.	Ref.	Author(s)	Application area	Prediction task	Model(s)	XAI method(s)	Performance metrics
Scopus	24	81	Li et al. (2026)	Carbon markets / China's national ETS / Carbon allowance pricing	One-day-ahead carbon allowance return and closing-price prediction	Random Walk (RW), Autoregressive Integrated Moving Average (ARIMA), Ridge Regression (Ridge), Least Absolute Shrinkage and Selection Operator Regression (LASSO), Elastic Net Regression (EN), K-Nearest Neighbors (KNN), Support Vector Regression (SVR), Random Forest (RF), Histogram-based Gradient Boosting (HGB), Multilayer Perceptron (MLP), Unweighted Machine Learning Ensemble (Ensemble)	SHAP (Global + Local), PFI (Global), PDP (Global)	RMSE, MAE, MAPE, DA
Scopus	25	82	Ma et al. (2026)	Carbon markets / Carbon pricing / Multiscale driver analysis	One-day-ahead carbon price forecasting in China's Hubei carbon market	Gated Recurrent Unit (GRU), Long Short-Term Memory (LSTM), Extreme Learning Machine (ELM); Empirical Mode Decomposition (EMD) hybrids: EMD–GRU, EMD–LSTM, EMD–ELM; Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) hybrids: CEEMDAN–GRU, CEEMDAN–LSTM, CEEMDAN–ELM; Sample Entropy (SE) reconstruction hybrids: CEEMDAN–SE–GRU, CEEMDAN–SE–LSTM, CEEMDAN–SE–ELM; Coefficient of Variation–Weighted Ensemble (VC ensemble), Varied Mean Absolute Percentage Error–Weighted Ensemble (VMAPE ensemble), Discounted Mean Squared Forecasting Error–Weighted Ensemble (DMSFE ensemble), Arithmetic Mean Ensemble (Mean ensemble), Cuckoo Search–Weighted Ensemble (CS ensemble)	SHAP (Global + Local)	SHAP (Global + Local)
Scopus	44	83	Sun et al. (2026)	Carbon markets / Regional ETS / Carbon pricing	Carbon price forecasting in Shanghai and Hubei across multiple horizons	Causal State-driven Network (CaStNet); Complete Ensemble Empirical Mode Decomposition with Adaptive Noise–Sample Entropy–Variational Mode Decomposition + Long Short-Term Memory–Transformer (CEEMDAN–SE–VMD + LSTM–Transformer); Artificial Lemming Algorithm–Long Short-Term Memory–Transformer (ALA–LSTM–Transformer); Informer; Decomposition Linear Model (DLinear); Long Short-Term Memory–Transformer (LSTM–Transformer); Temporal Convolutional Network (TCN); Convolutional Neural Network–Long Short-Term Memory (CNN–LSTM); Transformer; Gated Recurrent Unit (GRU); Extreme Gradient Boosting (XGBoost); Support Vector Regression (SVR); Autoregressive Integrated Moving Average (ARIMA)	SHAP (Global + Local + Interaction), LIME (Local)	RMSE, MAE, MAPE, R^2

Database	No.	Ref.	Author(s)	Application area	Prediction task	Model(s)	XAI method(s)	Performance metrics
Scopus	83	84	He et al. (2026)	Carbon markets / Carbon pricing / Risk monitoring	One-, five-, and ten-step-ahead carbon price forecasting	Support Vector Regression (SVR), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Bidirectional Gated Recurrent Unit (BiGRU), Temporal Convolutional Network (TCN); CEEMDAN–Convolutional Neural Network (CNN) hybrids: CEEMDAN–CNN–SVR, CEEMDAN–CNN–LSTM, CEEMDAN–CNN–GRU, CEEMDAN–CNN–BiGRU, CEEMDAN–CNN–TCN; attention hybrids: CEEMDAN–SVR–Attention, CEEMDAN–LSTM–Attention, CEEMDAN–GRU–Attention, CEEMDAN–BiGRU–Attention, CEEMDAN–TCN–Attention; combined hybrids: CEEMDAN–CNN–SVR–Attention, CEEMDAN–CNN–LSTM–Attention, CEEMDAN–CNN–GRU–Attention, CEEMDAN–CNN–BiGRU–Attention, CEEMDAN–CNN–TCN–Attention	SHAP (Global + Local), LIME (Local)	MAE, RMSE, MAPE, R^2
Scopus	171	85	Sayed et al. (2024)	Carbon markets / Carbon pricing / Investment decision support	Carbon price prediction from historical transaction features	Modified Light Spectrum Optimizer–Long Short-Term Memory (MLSO–LSTM); Long Short-Term Memory (LSTM); Modified Salp Swarm Algorithm–Long Short-Term Memory (MSSA–LSTM); Modified Whale Optimization Algorithm–Long Short-Term Memory (MWOA–LSTM); Modified African Vulture Optimization Algorithm–Long Short-Term Memory (MAVOA–LSTM); Modified Grey Wolf Optimizer–Long Short-Term Memory (MGWO–LSTM); Modified Differential Evolution–Long Short-Term Memory (MDE–LSTM); Random Forest Regressor (RF); Gradient Boosting Regressor (GB)	SHAP (Global + Local)	RMSE, R^2, MAE, MAPE, MSE
Scopus	183	86	Deng et al. (2024)	Carbon markets / Carbon pricing / Directional forecasting	One-, five-, and ten-step-ahead carbon price direction classification in Hubei	Multivariate Adaptive Regression Splines (MARS); Support Vector Machine (SVM); Random Forest (RF); Extreme Learning Machine (ELM); Extreme Gradient Boosting (XGBoost); Long Short-Term Memory (LSTM); Light Gradient Boosting Machine (LightGBM); Complete Ensemble Empirical Mode Decomposition with Adaptive Noise–LightGBM, Single-Factor Variant (CEEMDAN–LightGBM–S); CEEMDAN–Light Gradient Boosting Machine (CEEMDAN–LightGBM); CEEMDAN–Two-Stage Feature Selection–LightGBM (CEEMDAN–TFS–LightGBM); CEEMDAN–Bayesian Optimization Algorithm–LightGBM (CEEMDAN–BOA–LightGBM); CEEMDAN–Two-Stage Feature Selection–Bayesian Optimization Algorithm–LightGBM (CEEMDAN–TFS–BOA–LightGBM)	SHAP (Global + Local)	AUC, Accuracy, Precision, Recall, F1 score